

Indices - Brackets and Indices (Sheet 1)

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Simplify these by expanding first, then adding powers. Leave your answer in index form.

Example

$$4 + 4 + 4 = 12$$

$$(5^4)^3 = 5^4 \times 5^4 \times 5^4 = 5^{12}$$

1 $(7^3)^2 = \square \times \square = \square$

2 $(3^5)^3 = \square \times \square \times \square$
 $= \square$

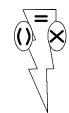
3 $(9^5)^4 = \square \times \square \times \square \times \square$
 $= \square$

Now letters - same process

4 $(y^2)^4 = \square \times \square \times \square \times \square$
 $= \square$

5 $(x^6)^5 = \square$
 $= \square$

6 $(k^3)^3 = \square$
 $= \square$



Brackets means you multiply the indexes. Simplify these. Leave answer in index form.

$$(m^6)^4 = m^{6 \times 4} = m^{24}$$

7 $(8^3)^4 = \square \times \square = \square$

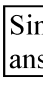
8 $(4^3)^5 = \square \times \square = \square$

9 $(y^2)^3 = \square \times \square = \square$

10 $(x^5)^3 = \square$

11 $(n^8)^2 = \square$

12 $(k^6)^5 = \square$



Simplify these. Leave your answer in index form.

13 $(8^2)^3 = \square$

14 $(5^4)^2 = \square$

15 $(y^7)^2 = \square$

16 $(k^3)^2 = \square$

17 $(m^3)^2 = \square$

18 $(a^6)^3 = \square$

19 $(10^4)^3 = \square$

20 $(h^5)^5 = \square$



Use your power key to evaluate these.

Example

$$5^4 = 625 \quad \text{Calc: } 5 \square x \square =$$

21 $2^4 = \square$ 22 $3^6 = \square$

23 $8^4 = \square$

24 $2^8 = \square$

25 $4^5 = \square$

26 $10^2 = \square$

27 $10^4 = \square$

28 $10^3 = \square$



Simplify these by expanding first, then multiplying. Use your calculator to calculate the numbers with powers.

Example

$$(2x^4)^3 = 2x^4 \times 2x^4 \times 2x^4 = 2^3 x^{12} = 8x^{12}$$

29 $(3h^5)^2 = \square \times \square = 3 \square h \square = \square$

30 $(5a^7)^3 = \square \times \square \times \square = 5 \square a \square = \square$

31 $(2n^6)^4 = \square \times \square \times \square \times \square = 2 \square n \square = \square$

32 $(3k^4)^3 = \square$

33 $(4c^5)^4 = \square$

Slide the power outside the brackets onto the number inside and multiply the two indexes for the power on the letter.

Example $(2x^4)^3 = 2^3 x^{12} = 8x^{12}$

34 $(3e^2)^3 = 3 \square e \square = \square$

35 $(5n^4)^2 = 5 \square n \square = \square$

36 $(2c^5)^4 = 2 \square c \square = \square$

37 $(3x^3)^4 = 3 \square x \square = \square$

38 $(6v^5)^3 = \square = \square$

39 $(10a^8)^2 = \square$

$= \square$

40 $(10y^5)^4 = \square$

$= \square$